**Multi-label Topic Detection in Hotel Reviews: System Description Paper NLP Shared Task**

Jeroen (Jozias) Kole – S0183512

## 1. Introduction

This paper describes a system developed for the NLP Shared Task on automatic topic detection in hotel reviews. The task is framed as multi-label classification (Tsoumakas & Katakis, 2009): given a short text snippet from a Booking.com review, the system must assign zero, one, or more topic labels from a fixed set of 20 labels such as room-cleanliness, staff, or catering-price.

The challenge is compounded by two factors. First, the label distribution is heavily skewed: common topics such as hotel-location and staff appear in hundreds of instances, while rare labels like bed-cleanliness occur in fewer than 20. Label imbalance of this kind is a well-documented challenge in multi-label classification, where models trained without correction tend to systematically ignore rare labels in favour of frequent ones (Chalkidis et al., 2020; Fehle et al., 2023). Second, the test data introduces two new languages (Dutch and Italian) that are absent from the training corpora.

Given these constraints, the primary design goals were: (1) cross-lingual transfer capability, and (2) robust handling of label imbalance at both training time (through class weighting and synthetic augmentation) and inference time (through per-label threshold tuning). A secondary goal was to empirically assess whether per-label threshold tuning alone is sufficient to handle label imbalance, or whether additional synthetic data augmentation provides further gains.

## 2. Dataset

### 2.1 Provided training data

The training corpus consists of 1,000 annotated Booking.com hotel reviews across five languages (German, Spanish, French, English, and a second Spanish subcorpus from Mexico City). Each review is split into a liked and a disliked section, yielding 2,000 text instances, annotated with 20 topic labels following an aspect-attribute structure (e.g. room-cleanliness, bathroom-facilities). Each instance is treated as an independent text instance with its own label set. Labels were converted to binary vectors of length 20; unlabelled instances were retained as informative negatives.

### 2.2 Synthetic augmentation for unseen languages

To provide domain- and task-specific fine-tuning signal in Dutch and Italian, 200 synthetic hotel reviews per language were generated using the Claude Sonnet 4.5 API. Labels were sampled from empirically derived per-label probabilities to match the training distribution (mean 2.11 labels per instance, zero-label rate 3.4%), then passed as explicit assignments in the prompt. Few-shot seeds were drawn from the closest related training language (German

for Dutch, Spanish for Italian). Temperature was set to 0.7 (see Appendix A.1). Typicality scores (Zhang, 1992; Van Nooten & Daelemans, 2023) against the reference language corpora were 1.45 ± 0.34 for Dutch and 1.52 ± 0.36 for Italian (see Appendix A.3 for synthetic data quality).

### 2.3 Targeted augmentation for rare labels

Six labels were severely underrepresented in the training data ($n < 60$) and were upsampled to a minimum of 100 instances using the same API, attempting to reduce the imbalance while keeping the synthetic share of the corpus modest. The distributional constraints governing generation are detailed in Appendix A.2. The augmentation produced 199 synthetic reviews (313 text instances, 756 label instances, per-label counts in Table A1). Typicality of the augmented instances against the real training corpus was 1.79 ± 0.32, indicating that synthetic instances are similar to real instances of the same label (see Appendix A.3).

## 3. Methodology

### 3.1 Experimental setup and model selection

Seven systems were evaluated on the same stratified validation set (10% of real training data, held out via iterative stratification; Sechidis et al., 2011) to ensure comparability (see Table 1). System 1 is a TF-IDF + Logistic Regression baseline. Systems 2–7 fine-tune XLM-RoBERTa-base under an incremental ablation design: System 2 is the transformer baseline trained on real data only; System 3 adds per-label threshold tuning; System 4 adds synthetic NL/IT data; System 5 combines NL/IT augmentation with threshold tuning and is the final submitted system; Systems 6 and 7 further add targeted rare-label augmentation, without and with threshold tuning respectively. To assess result stability, the best performing system was additionally retrained with three random seeds (42, 123, 456).

### 3.2 Model architecture

XLM-RoBERTa-base (Conneau et al., 2020) was selected as the final architecture: pretrained on 2.5TB of text spanning 100 languages including Dutch and Italian, it is well-suited to the multilingual training data and the unseen test languages. A linear classification head with 20 outputs was added on top of the [CLS] representation and fine-tuned end-to-end on the combined training set. Input sequences were tokenized using the XLM-RoBERTa sentencepiece tokenizer, truncated or padded to a maximum length of 128 tokens. The model was optimised using AdamW with batch size 16, early stopping with patience 3, and mixed-precision (fp16) training on GPU. Hyperparameters were selected via grid search over learning rates $\in$ {1e-5, 2e-5, 3e-5} and warmup ratios $\in$ {0.06, 0.1, 0.2} using macro F1 on the validation set. Due to computational costs, the grid search was conducted on the System 4 configuration (real + synthetic NL/IT, no threshold tuning) as a representative setting; the identified hyperparameters were then applied to all systems. The best configuration was lr=3e-5, warmup=0.06 (Appendix A4). Training ran for up to 10 epochs with weight decay 0.01.

### 3.3 Loss function and class weighting

Each label is treated as an independent binary classification problem, using Binary Cross-Entropy loss (Tsoumakas & Katakis, 2009; Fehle et al., 2023). To counter label imbalance, per-label positive class weights are computed as the ratio of negative to positive training instances, capped at 30, penalising missed predictions on rare labels more heavily. Class weighting and targeted augmentation address label imbalance at complementary levels: the former reweights the loss, the latter increases raw positive counts.

### 3.4 Per-label threshold tuning

The default decision threshold of 0.5 is not necessarily optimal per label (Lipton et al., 2014). Thresholds are tuned post-hoc on the validation set by searching [0.10, 0.90] in steps of 0.05 and selecting the value maximising per-label F1. This adds no training overhead and directly optimises the evaluation metric, making it a low-cost approach to handling label imbalance at inference time. The saved thresholds will be applied to the test set.

## 4. Preliminary performance metrics

Table 1 reports macro F1 across all seven systems on the validation set. The TF-IDF + Logistic Regression baseline achieves a macro F1 of 0.425, confirming that sparse representations are insufficient for this multilingual task. Among transformer systems, per-label threshold tuning is the single most effective intervention, adding between 0.09 and 0.13 macro F1 in every configuration. Adding synthetic NL/IT data (System 4 vs System 2: 0.614 vs 0.534) yields a consistent improvement of 0.08 on the validation set. Since the validation set contains none of the target test languages, this gain likely reflects improved cross-lingual generalisation. Targeted rare-label augmentation (System 6 vs System 4: 0.611 vs 0.614) produces no improvement. A post-hoc hyperparameter search on the System 6 configuration confirms that hyperparameter mismatch (as the original search was only conducted in system 4) is not the cause: the optimal learning rate is identical (3e-5) and the applied warmup ratio (0.06) is within 0.010 macro F1 of the optimum (warmup=0.10, F1=0.614 vs 0.603). The high typicality of the augmented instances suggests that generation quality is also not the problem. The more fundamental constraint is the absolute scarcity of real examples. With as few as 29 instances of bed-cleanliness before augmentation, the few-shot seeds passed to the generator might themselves be too sparse and homogeneous to produce genuinely varied output, particularly across multiple languages, regardless of how many synthetic instances are added. A related constraint is that RLHF alignment causes models to favour familiar, prototypical outputs, driven by typicality bias in human preference data (Zhang et al., 2025). Even with sufficient synthetic volume, generated instances may therefore cluster around the same prototypical descriptions and add little new discriminative signal. Beyond this, hotel aspect labels are semantically very close, and it may simply be difficult for a generative model to produce short review texts that reliably distinguish between fine-grained categories like bed-cleanliness and bathroom-general. This is consistent with the generalisation experiment in Appendix A.5, where augmented data trained in isolation underperforms an equal-sized real sample (0.193 vs 0.224).

The final submitted system (System 5) achieves a mean macro F1 of 0.710 ± 0.005 across three random seeds (42, 123, 456), indicating stable performance. Per-label F1 scores are reported in Table A7.

| System | Macro F1 |
|---|---|
| 1. TF-IDF + Logistic Regression | 0.425 |
| 2. XLM-R | 0.534 |
| 3. XLM-R + threshold tuning | 0.671 |
| 4. XLM-R + NL/IT | 0.614 |
| **5. XLM-R + NL/IT + threshold tuning[1]** | **0.714** |
| 6. XLM-R + NL/IT + rare-label aug | 0.611 |
| 7. XLM-R + NL/IT + rare-label aug + threshold tuning | 0.703 |

Table 1: Macro F1 on the validation set across all seven systems.

## 5. Limitations

Per-label thresholds were optimised and evaluated on the same 200-instance validation split, risking overfitting particularly for low-support labels (e.g. bed-cleanliness, n=2); the reported macro F1 of 0.710 is therefore a likely overestimate of test set performance. Rare labels with very low support consequently remain the primary challenge despite augmentation and class weighting. The modest synthetic share of the training corpus (14%) may have limited the impact of rare-label augmentation, though the generalisation results in Appendix A.5 suggest the bottleneck is signal quality rather than volume. Finally, hyperparameters were selected via grid search on System 4 only, meaning ablation results may slightly underestimate other configurations. Computational constraints also precluded cross-validation for more stable threshold estimates and the use of XLM-RoBERTa-large, which would likely improve cross-lingual transfer.

## 6. Conclusion

This paper described a multi-label classification system for hotel review topic detection based on fine-tuned XLM-RoBERTa-base. The ablation results point to two clear conclusions. First, per-label threshold tuning is the most cost-effective intervention, adding no training overhead while consistently yielding the largest macro F1 gains. Second, targeted rare-label augmentation produced no measurable gain in any configuration, suggesting that class weighting and threshold tuning outperform synthetic augmentation as strategies for handling residual label imbalance for this task. Whether the cross-lingual gains from synthetic NL/IT augmentation observed on the validation set transfer to the Dutch and Italian test subcorpora remains the primary open question.

[1] System 5 was selected as the final submitted system. Multi-seed evaluation across three random seeds (42, 123, 456) yields a macro F1 of 0.710 ± 0.005, confirming stable performance.

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## Appendix

### A.1 Cross-lingual augmentation (NL/IT)

Reviews were generated in batches of 10 via the Claude Sonnet 4.5 API. Each prompt contained a system instruction specifying the task format, four randomly resampled few-shot examples from the training corpus, and pre-sampled label assignments. Few-shot seeds were drawn from the closest related language (German for Dutch, Spanish (ES/MX) for Italian). Temperature was set to 0.7 to balance consistency and creativity. Although this exceeds the 0.6 used by Van Nooten & Daelemans (2023), who found higher values produced ungrammatical output with GPT-3.5, manual inspection of a sample of generated reviews confirmed grammatical fluency in both Dutch and Italian, likely reflecting the improved language quality of more recent generative models. Label fields were overridden post-generation with the pre-sampled targets to prevent model label bias.

System prompt:

"*You are a hotel review generator for an NLP dataset. Your task is to write realistic multilingual Booking.com reviews that precisely match the topic label assignments provided. Respond ONLY with a valid JSON array, no preamble, no markdown fences*."

User prompt template:

"*Generate {batch_size} hotel reviews in {language} for hotels in {city}.*

*## Style examples: The following are real Booking.com reviews. Match their tone, register, and length variation when writing your reviews. Note how topics are signalled naturally in the text: {four_few_shot_examples}*

*## Your task: Now generate {batch_size} new reviews in {language}. For each review, you are given the EXACT topic labels that the text must cover. Write content that naturally reflects those topics — and only those topics. If a label list is empty, write a short generic sentence.*

*Label assignments (index matches output array position): {label_assignments}*

*## Rules*

*— Write in {language} as a real guest would*

*— do not translate the examples above*

*— Match the style variation in the examples: some short, some longer*

*— "disliked" can be an empty string when disliked_labels is empty*

*— Do NOT include "Liked:" or "Disliked:" prefixes*

*Return a JSON array of {batch_size} objects with keys: liked, topic_detection_liked, disliked, topic_detection_disliked. topic_detection fields must contain EXACTLY the assigned labels, semicolon-separated, or empty string*."

### A.2 Rare label augmentation

Generation preserved four properties of the real corpus. First, the balance between liked and disliked reviews was maintained in proportion to the empirical distribution observed for each label, with a minimum of 30% for each of the liked and disliked sections. Second, the language distribution of the synthetic reviews reflected the real per-label language frequencies found in the dataset. Third, the augmentation preserved the multilabel co-occurrence structure of the corpus. The number of labels assigned to each synthetic instance was sampled from the empirical distribution of the training data (mean = 2.18; median = 2), while highly frequent labels (n > 200) were down-weighted by a factor of 0.3 to avoid overrepresentation. Fourth, the paired liked/disliked row structure of the original training data was retained in the generated output.

In total, the augmentation process produced 199 synthetic reviews, corresponding to 313 text instances and 756 additional label instances; multilabel co-occurrence preservation resulted in incidental augmentation of more common labels (Table A1).

Generation was performed in batches, with each batch targeting one specific rare label. The prompting strategy depended on whether the generated instance was single-label or multilabel. For single-label instances it instructed the model to focus exclusively on the target label; for multi-label instances it specified the total label count and listed preferred co-occurring labels.

User prompt template:

"*You are generating synthetic hotel review excerpts for a multilingual NLP dataset.*

*Generate {n} new hotel review text instances in {language}. Each instance should be {positive/negative}. Each must be relevant to the topic: "{target_label}".*

*Label rules: {multilabel_instruction}*

*Available labels: {full_label_list}*

*Output ONLY a JSON array. Each element must have:*

*— "text": the review excerpt (realistic, natural, varied length, in {language})*

*— "labels": array of applicable label strings from the available labels list*

*The label "{target_label}" MUST appear in every generated instance's labels. Do not include labels that are not in the available labels list. Do not include the prefix 'Liked ·' or 'Disliked ·' in the text.*

*Here are real examples to guide style and label co-occurrence: {up_to_three_few_shot_examples}*

*Return only the JSON array, no other text.*"

| Label | Before | Added | After |
|---|---|---|---|
| bed-cleanliness | 29 | 71 | 100 |
| bathroom-general | 35 | 71 | 106 |
| catering-price | 50 | 52 | 102 |
| bathroom-cleanliness | 56 | 88 | 144 |
| bathroom-atmosphere | 58 | 51 | 109 |
| parking | 59 | 41 | 100 |
| hotel-cleanliness | 128 | 49 | 177 |
| room-cleanliness | 130 | 59 | 189 |
| hotel-price | 133 | 52 | 185 |
| room-general | 148 | 41 | 189 |
| bed-general | 175 | 5 | 180 |
| bathroom-facilities | 174 | 70 | 244 |
| hotel-facilities | 175 | 12 | 187 |
| hotel-general | 241 | 18 | 259 |
| hotel-atmosphere | 220 | 4 | 224 |
| room-facilities | 261 | 14 | 275 |
| room-atmosphere | 318 | 14 | 332 |
| catering-general | 230 | 21 | 251 |
| hotel-location | 479 | 0 | 479 |
| staff | 648 | 23 | 671 |
| **Total instances** | **2000** | **756** | **2756** |

Table A1: Label counts before and after targeted augmentation (sorted by pre-augmentation frequency).

### A.3 Synthetic data quality

Synthetic NL/IT texts are shorter and substantially more uniform in length than real data (see Table A2), a known limitation of LLM-generated reviews (Van Nooten & Daelemans, 2023). The augmented rare-label data shows greater length variation than the synthetic NL/IT data, likely reflecting the wider range of topics and languages covered.

| Dataset | Mean length | Std | N |
|---|---|---|---|
| Real (DE/ES/FR/GB/MX) | 23.6 | 38.1 | 2000 |
| NL synthetic | 18.0 | 9.3 | 400 |
| IT synthetic | 15.5 | 8.6 | 400 |
| Augmented (rare labels) | 19.4 | 12.4 | 398 |

Table A2: Mean text length per dataset.

Typicality (Zhang, 1992) measures how well a generated instance represents its assigned label compared to other labels. In other words, whether a synthetic text about bed-cleanliness actually looks more like real bed-cleanliness texts than like texts about, for example, hotel-location. It is computed using the paraphrase-multilingual-mpnet-base-v2 sentence encoder: for each synthetic instance, the average cosine similarity to same-label reference instances is divided by the average cosine similarity to different-label instances. Scores above 1.0 indicate label-coherent generation. For the cross-lingual datasets, NL synthetic instances were compared against the German subcorpus and IT synthetic instances against the Spanish subcorpora (ES/MX), as the closest related training languages.

| Dataset | Reference | Typicality (mean ± std) | N |
|---|---|---|---|
| NL synthetic | German (DE) | 1.45 ± 0.34 | 386 |
| IT synthetic | Spanish (ES/MX) | 1.52 ± 0.36 | 386 |
| Augmented (rare labels) | Real corpus (all) | 1.79 ± 0.32 | 313 |

Table A3: Typicality scores for synthetic datasets.

All datasets score above 1.0, confirming label-coherent generation. Scores for the cross-lingual datasets (NL, IT) should be interpreted with caution, as they partly reflect linguistic similarity between related languages rather than label coherence alone. For reference, Van Nooten & Daelemans (2023) report 1.2 for GPT-3.5 in a same-language setting.

### A.4 Hyperparameter Search Results

Grid search was conducted over learning rates ∈ {1e-5, 2e-5, 3e-5} and warmup ratios ∈ {0.06, 0.1, 0.2} on the System 4 configuration (real + synthetic NL/IT data), evaluated on the real validation set (n=200). Results are reported in Table A4.

| Learning rate | Warmup ratio | Macro F1 |
|---|---|---|
| 1e-5 | 0.06 | 0.438 |
| 1e-5 | 0.10 | 0.445 |
| 1e-5 | 0.20 | 0.442 |
| 2e-5 | 0.06 | 0.514 |
| 2e-5 | 0.10 | 0.523 |
| 2e-5 | 0.20 | 0.530 |
| **3e-5** | **0.06** | **0.555** |
| 3e-5 | 0.10 | 0.555 |
| 3e-5 | 0.20 | 0.553 |

Table A4: Macro F1 on the validation set for each hyperparameter combination.

Learning rate is the dominant factor; warmup ratio has negligible effect at lr=3e-5 (range: 0.553–0.555). The best configuration (lr=3e-5, warmup=0.06) was applied to all systems.

| Learning rate | Warmup ratio | Macro F1 |
|---|---|---|
| 1e-5 | 0.06 | 0.505 |
| 1e-5 | 0.10 | 0.511 |
| 1e-5 | 0.20 | 0.513 |
| 2e-5 | 0.06 | 0.572 |
| 2e-5 | 0.10 | 0.575 |
| 2e-5 | 0.20 | 0.571 |
| **3e-5** | **0.10** | **0.614** |
| 3e-5 | 0.06 | 0.603 |
| 3e-5 | 0.20 | 0.599 |

Table A5: System 6 HP search: macro F1 on the validation set (real + synthetic NL/IT + rare-label augmentation training data).

The optimal learning rate (3e-5) is consistent with Table A4. The warmup ratio preference inverts marginally (0.10 > 0.06), but the absolute gap is 0.010 macro F1. The applied configuration (lr=3e-5, warmup=0.06) was therefore near-optimal for Systems 6/7.

### A.5 Generalisation of synthetic data

To assess whether each synthetic dataset encodes transferable task signal before committing it to training, models were trained on each synthetic dataset alone and evaluated on the real validation set (cf. Van Nooten & Daelemans, 2023). Results are reported in Table A6.

| Training data | Macro F1 |
|---|---|
| Real train (full, n=1800) | 0.374 |
| Real sample (n=400) | 0.230 |
| NL synthetic only (n=400) | 0.189 |
| IT synthetic only (n=400) | 0.243 |
| Real sample (n=398) | 0.224 |
| Augmented only (n=398) | 0.193 |

Table A6: Generalisation results: trained on synthetic data only, tested on real validation (n=200).

All synthetic datasets perform above chance, confirming that generated text encodes task-relevant signal. These results should be interpreted as a plausibility check rather than a validation of cross-lingual transfer: training on synthetic NL/IT data and testing on a real validation set drawn from the original five languages does not replicate the actual inference condition, and whether the augmentation improves performance on unseen Dutch and Italian test instances remains an open empirical question.

Italian synthetic data (0.243) outperforms an equal-sized real sample (0.230), consistent with its high typicality score and the relative closeness of Italian to the Spanish training subcorpora. Dutch synthetic data (0.189) falls below the real sample baseline, possibly reflecting greater typological distance from the training languages and warranting caution about the expected cross-lingual gain for Dutch specifically. Augmented rare-label data (0.193) similarly underperforms its real sample counterpart (0.224), consistent with its negligible downstream impact in the main ablation. The limitations of the synthetic augmentation are discussed in the paper.

**A.6 Per-label F1-scores**

Per-label F1 scores vary substantially across the 20 labels, largely tracking validation support. High-frequency labels such as staff (F1 0.862, n=65) and hotel-location (F1 0.975, n=61) are classified reliably, while low-support labels show the weakest performance. bathroom-atmosphere (F1 0.471, n=6) and bed-cleanliness (F1 0.500, n=2) are the most difficult labels; the latter has only 2 positive instances in the validation set, making its F1 score statistically unreliable and likely an overestimate given that threshold tuning was performed on the same split.

| Label | F1 | Support |
|---|---|---|
| hotel-general | 0.627 | 27 |
| hotel-atmosphere | 0.677 | 30 |
| hotel-cleanliness | 0.720 | 14 |

| Label | F1 | Support |
|---|---|---|
| hotel-facilities | 0.625 | 20 |
| hotel-location | 0.975 | 61 |
| hotel-price | 0.750 | 15 |
| room-general | 0.622 | 17 |
| room-atmosphere | 0.778 | 40 |
| room-cleanliness | 0.600 | 14 |
| room-facilities | 0.690 | 33 |
| bathroom-general | 0.571 | 4 |
| bathroom-atmosphere | 0.471 | 6 |
| bathroom-cleanliness | 0.625 | 6 |
| bathroom-facilities | 0.750 | 15 |
| bed-general | 0.913 | 22 |
| bed-cleanliness | 0.500 | 2 |
| catering-general | 0.893 | 27 |
| catering-price | 0.800 | 5 |
| parking | 0.833 | 6 |
| staff | 0.862 | 65 |

Table A7: Per-label F1-score for the final submitted system (System 5: XLM-R + NL/IT + threshold tuning), evaluated on the validation set.